

NLP Assignment 1 (10%)

For the below dataset, build the following models to identify if a tweet is positive, negative or neutral using the following models

1. SVM Model using GloVe or FastText or ELMo
2. Deep Unidirectional RNN Model using TF-IDF with unigram+bigram
3. Deep Bidirectional GRU Model with an embedding layer to learn embeddings
4. Single Layer Unidirectional LSTM Model using word2vec (CBOW or Skipgram and any vector size from 64 to 300)
5. Deep Bidirectional LSTM Model using an embedding layer to learn embeddings

Make sure to tune your models extensively and explain your rationale for the tuning approaches used each iteration

Do call out what else could be done to tune the model and how it would have helped (w/ some numbers) at the top/bottom of your notebook.

Please make sure there are enough data points in the test set (>5000) for Confusion Matrix, AUC etc.

Submit your python notebook run sequentially end to end as a .pdf file

At the beginning of the assessment please submit the following table. Please fill the result values in the table **accurately** else you will not get marks for the corresponding task (This table only shows for 2 models but you need to add similarly for all 5 models)

Task	Status	Results	Individual Responsible
SVM Model using GloVe, Fasttext or ELMo	Config of the model trained	Train Time = ?	
Training Data Check	Confusion Matrix Built?	F1 Score for Positive = ? F1 Score for Negative = ?	

Task	Status	Results	Individual Responsible
Feature Engg	AUC plotted?	AUC = ?	
	Accuracy computed?	Accuracy = ?	
	Categorical/Numerical Features Added?	Name 2 features added?	
		AUC = ?	
Model Tuning Iteration 1	Observations? What was going wrong? Changes Made?	F1 Score for Positive = ? F1 Score for Negative = ?	
Model Tuning Iteration 2	Observations? What was going wrong? Changes Made?	AUC = ? F1 Score for Positive = ? F1 Score for Negative = ?	
Interpretability	Interpretability Implemented? Local or Global?	2 interesting findings?	
Testing Data Check	Confusion Matrix Built?	F1 Score for Positive = ? F1 Score for Negative = ?	
Next Steps	AUC plotted?	AUC = ?	
	Accuracy computed?	Accuracy = ?	
	List out 2-3 possible next steps for the this model		
Deep Unidirectional RNN Model with an embedding layer	Type of Smoothing Used? Config of model trained?	Train Time = ?	
Training Data Check	Confusion Matrix Built?	F1 Score for Positive = ?	

			F1 Score for Negative = ?
		AUC plotted?	AUC = ?
		Accuracy computed?	Accuracy = ?
Feature Engg		Feature Weightages Added?	2 Features with the Highest Weights?
			AUC = ?
Model Tuning Iteration 1		Observations? What was going wrong? Changes Made?	F1 Score for Positive = ? F1 Score for Negative = ?
			AUC = ?
Model Tuning Iteration 2		Observations? What was going wrong? Changes Made?	F1 Score for Positive = ? F1 Score for Negative = ?
			AUC = ?
Testing Data Check	Confusion Matrix Built?		F1 Score for Positive = ? F1 Score for Negative = ?
		AUC plotted?	AUC = ?
		Accuracy computed?	Accuracy = ?
Next Steps	List out 2-3 possible next steps for Naive Bayes Models		
Data Preprocessing and Feature Engg	Regex Used?	# of regex used?	
	Emoji Handling Done?	# of emojis handled?	
	Slang Handling Done?	# of slangs handled?	

Abbreviations
Handled?

of abbreviations handled?

Class Separability
Checked?

Featureset with best separability?

Train and Test
Handled Correctly?

Data Preprocessing and Feature Engg: Steps
taken before train test split and steps taken
after train test split?